

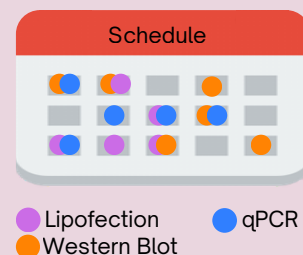
Cellular Research Track

About the Program

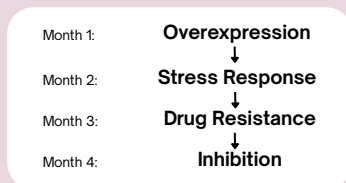
Cellular Discovery Track is the first fully flexible research program that allows you to completely customize your learning path: choosing your own hours and the laboratory experimentation you wish to complete. Designed to cultivate researchers prepared for the biotech landscape, participants generate professional scientific data and publication-quality figures while investigating real medical challenges. These findings are ultimately synthesized into a collaborative research manuscript, earning participants verified technical badges and co-authorship recognition. Across a central theme, each of the 4-months explores progressively deeper layers of the same biological pathway, building a comprehensive understanding of how the system operates.

Flexible Research Model

In-person laboratory sessions are fully modular: you customize your own track and book only the hands-on experiments you choose to perform. You only pay for the laboratory work you add to your portfolio, while remaining free to explore additional topics through the online course. Every participant receives full access to our comprehensive online curriculum, enabling independent mastery of the scientific concepts and protocols at your own pace.



★ Do stress-activated survival pathways enable cells to adapt and resist chemotherapy?



Hypothesis-Driven Investigation

Within a central hypothesis, participants address a focused research aim each month using advanced biotechnology techniques to generate measurable evidence, progressively uncovering the mechanistic connections within the pathway.

Colaborative Research Manuscript

Students learn to convert experimental data into publication-quality figures, developing skills in quantitative analysis and scientific communication. These figures are compiled into a collaborative research paper answering the central hypothesis. Authorship is designated according to meaningful contribution, including data collection, experimental work, and intellectual synthesis of results.

Eligibility

All students **MUST**:

- Be over the age of 18
- Complete our laboratory safety certification online course (\$30)
- Complete our in person laboratory orientation
- Submit signed user agreement and code of conduct

OpenLabs maintains a zero tolerance policy for sanitation and safety violations. Failure to follow these protocols results in immediate user review or expulsion.

Research Training Pathway

Each in-person laboratory module is structured to advance the central hypothesis through a distinct experimental layer, contributing evidence that builds toward the overarching collaborative research paper. Conceptual understanding also extends beyond the bench through the free online Research Track courses, which prepare students to perform protocols confidently, strengthen data interpretation skills, and provide tools for independent scientific exploration.



Online Courses:



- Recombinant Gene Design
- Plasmid Design

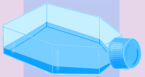
Module 1: Bacterial Culture

\$35.00

Develop the skills to propagate, screen, and purify engineered plasmids through professional-grade bacterial transformation workflows.

Experimental Procedures:

- Bacterial Transformation
- Bacterial Streaking and Plating
- Colony Screening
- Plasmid MiniPrep



- Lipid Nanoparticles
- Electroporation
- CRISPR

Module 2: Cell Culture & Transfection

\$45.00

Train in sterile mammalian cell culture and gene delivery techniques to manipulate cellular behavior in a controlled laboratory environment.

Experimental Procedures:

- Cell Seeding and Passaging
- Hemocytometry
- Cell Viability Assays
- Transient Transfection through Lipofection



- PCR Primer Design
- Analyzing C-values

Module 3: Gene Quantification

\$35.00

Decode how cells respond at the transcriptional level by measuring precise changes in gene expression to test and validate biological hypotheses.

Experimental Procedures:

- Phenol-chloroform RNA isolation
- PCR amplification
- RTPCR and qPCR



- Recombinant Protein Design
- AlphaFold
- Docking Models
- Binding Affinity

Module 4: Protein Quantification

\$45.00

Demonstrate whether molecular pathways are functionally engaged by measuring changes in protein abundance that drive cellular behavior.

Experimental Procedures:

- Protein Extraction
- BCA Assay
- Westernblot

Central Hypothesis:

Do stress-activated survival pathways enable cells to adapt and resist chemotherapy?



Construct stress survival expression vectors



Overexpress in cells



Quantify stress activation and resistance gene upregulation



Validate functional adaption through chemoresistance proteins.